Name: Theodore AL Sammour

Subject: EECE 503Q

Lab: 007

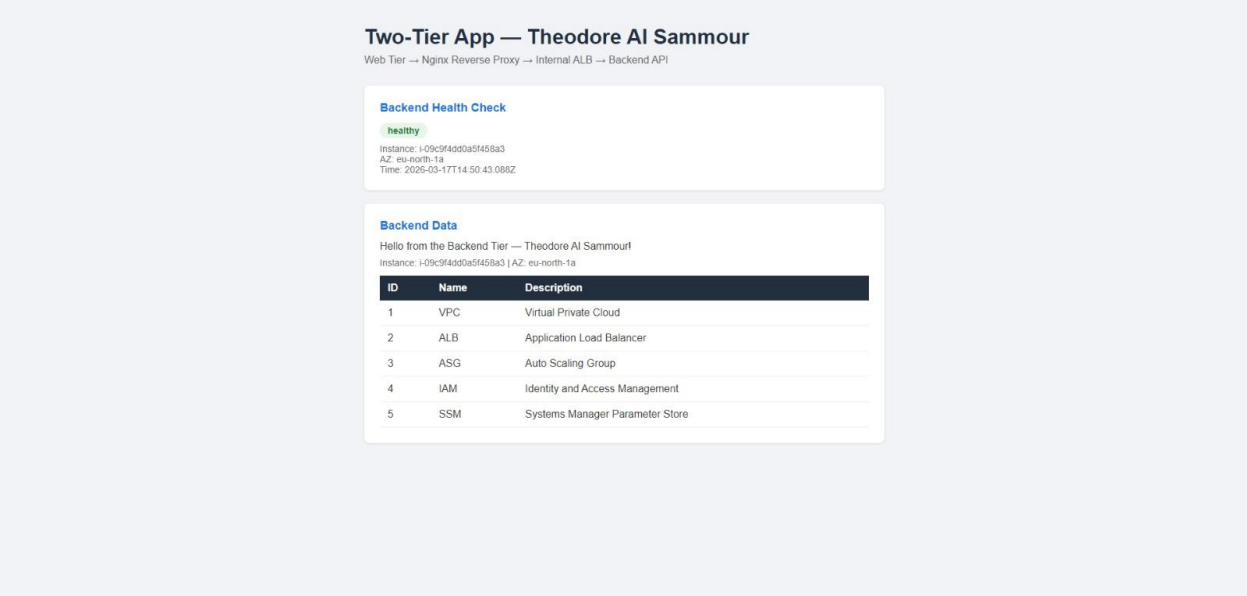
Github: https://github.com/TheodoreALSammour/Lab-007---Theodore-AL-Sammour.git

```
aws_lb_listener.internal: Creating...
aws_launch_template.web: Creating...
aws_lb_listener.internal: Creation complete after 1s [id=arn:aws:elasticloadbalancing:eu-north-1:505999908978:l
istener/app/lab007-internal-alb/e589738bc0ae1669/808ad68fda56789c]
aws_lb.external: Creation complete after 2m14s [id=arn:aws:elasticloadbalancing:eu-north-1:505999908978:loadbal
ancer/app/lab007-external-alb/13e051b792cd9a0d]
aws_lb_listener.external: Creating...
aws_lb_listener.external: Creation complete after 1s [id=arn:aws:elasticloadbalancing:eu-north-1:505999908978:l
istener/app/lab007-external-alb/13e051b792cd9a0d/6a7910bda8115a04]
aws_launch_template.web: Creation complete after 7s [id=lt-0a7afa902ca9d409c]
aws_autoscaling_group.web: Creating...
aws_autoscaling_group.web: Still creating... [00m10s elapsed]
aws_autoscaling_group.web: Creation complete after 18s [id=lab007-web-asg]
aws_autoscaling_attachment.web_attach: Creating...
aws_autoscaling_attachment.web_attach: Creation complete after 0s [id=lab007-web-asg-20260317143053048500000000a
]

Apply complete! Resources: 37 added, 0 changed, 0 destroyed.

Outputs:

bastion_public_ip = "13.63.160.6"
external_alb_dns = "lab007-external-alb-1112607047.eu-north-1.elb.amazonaws.com"
internal_alb_dns = "internal-lab007-internal-alb-1156372751.eu-north-1.elb.amazonaws.com"
ssm_parameter_paths = [
  "/lab007/db_password",
  "/lab007/app_env",
]
vpc_id = "vpc-0a47a0eca5ffee33e"
```



Two-Tier App — Theodore Al Sammour

Web Tier → Nginx Reverse Proxy → Internal ALB → Backend API

Backend Health Check

healthy

Instance: i-0534998f54e43f946
AZ: eu-north-1b
Time: 2026-03-17T14:51:56.167Z

Backend Data

Hello from the Backend Tier — Theodore Al Sammourl

Instance: i-0534998f54e43f946 | AZ: eu-north-1b

ID	Name	Description
1	VPC	Virtual Private Cloud
2	ALB	Application Load Balancer
3	ASG	Auto Scaling Group
4	IAM	Identity and Access Management
5	SSM	Systems Manager Parameter Store

lab007-web	i-01c7466ecb24e7bee	Running		t3.micro
lab007-bastion	i-0d12e4ee8d05d9a6c	Running		t3.micro
lab007-backend	i-0778c9865c6874043	Running		t3.micro

lab007-backend-tg	arn:aws:elasticloadbalancin...	3000	HTTP	Instance	lab007-internal-alb	vpc-0a47a0eca5ffee33e
lab007-web-tg	arn:aws:elasticloadbalancin...	80	HTTP	Instance	lab007-external-alb	vpc-0a47a0eca5ffee33e

Target group: lab007-backend-tg

Details | Targets | Monitoring | Health checks | Attributes | Tags

Details

arn:aws:elasticloadbalancing:eu-north-1:505999908978:targetgroup/lab007-backend-tg/02a7a3dad248cc75

Target type	Protocol : Port	Protocol version	VPC
Instance	HTTP: 3000	HTTP1	vpc-0a47a0eca5ffee33e
IP address type	**Load balancer**		
IPv4	lab007-internal-alb		

2	2	0	0	0	0
Total targets	Healthy	Unhealthy	Unused	Initial	Draining
	0 Anomalous				

▸ **Distribution of targets by Availability Zone (AZ)**

Select values in this table to see corresponding filters applied to the Registered targets table below.

lab007-web-tg	arn:aws:elasticloadbalancin...	80	HTTP	Instance	lab007-external-alb	vpc-0a47a0eca5ffee33e

Target group: lab007-web-tg

Details | Targets | Monitoring | Health checks | Attributes | Tags

Details

arn:aws:elasticloadbalancing:eu-north-1:505999908978:targetgroup/lab007-web-tg/c26fcb5c29f222ae

Target type	Protocol : Port	Protocol version	VPC
Instance	HTTP: 80	HTTP1	vpc-0a47a0eca5ffee33e
IP address type	**Load balancer**		
IPv4	lab007-external-alb		

2	2	0	0	0	0
Total targets	Healthy	Unhealthy	Unused	Initial	Draining
	0 Anomalous				

▸ **Distribution of targets by Availability Zone (AZ)**

Select values in this table to see corresponding filters applied to the Registered targets table below.

vpc-0a47a0eca5ffee33e / lab007-vpc

Details | Resource map | CIDRs | Flow logs | Tags | Integrations

Resource map Info — Show all details

VPC
Your AWS virtual network
lab007-vpc
10.1.0.0/16
No IPv6

Subnets (4)
Subnets within this VPC
eu-north-1a
A lab007-public-1
10.1.1.0/24
No IPv6
A lab007-private-1
10.1.3.0/24
No IPv6
eu-north-1b
B lab007-public-2
10.1.2.0/24
No IPv6
B lab007-private-2
10.1.4.0/24
No IPv6

Route tables (3)
Route network traffic to resources
lab007-public-rt
2 subnet associations
2 routes including local
rtb-02ab33918bd8d9fca
No subnet associations
1 route including local
lab007-private-rt
2 subnet associations
2 routes including local

Network Connections (2)
Connections to other networks
lab007-igw
Internet routes to 2 public subnets
2 private subnets route to the Internet
lab007-nat
Public NAT gateway
1 ENI with 1 EIP

Parameter Store

My parameters | Public parameters | Settings

You have upgrades available. Manage settings to increase your capacity. — Manage settings

My parameters — View details | Edit | Delete | Create parameter

Search — 1

Name	Tier	Type	Last modified
/lab007/app_env	Standard	String	Tue, 17 Mar 2026 14:28:21 GMT
/lab007/db_password	Standard	SecureString	Tue, 17 Mar 2026 14:28:21 GMT

```
[ec2-user@ip-10-1-3-50 ~]$ curl http://localhost:3000/api/health
{"status":"healthy","tier":"backend","name":"Theodore AL Sammour","instanceId":"i-06ca0d91ba0abe335","availabilityZone"
"eu-north-1a","timestamp":"2026-03-17T20:23:43.865Z"}[ec2-user@ip-10-1-3-50 ~]$
```

```
Plan: 0 to add, 0 to change, 37 to destroy.

Changes to Outputs:
  - bastion_public_ip   = "13.63.160.6" -> null
  - external_alb_dns    = "lab007-external-alb-1112607047.eu-north-1.elb.amazonaws.com" -> null
  - internal_alb_dns    = "internal-lab007-internal-alb-1156372751.eu-north-1.elb.amazonaws.com" -> null
  - ssm_parameter_paths = [
      - "/lab007/db_password",
      - "/lab007/app_env",
    ] -> null
  - vpc_id              = "vpc-0a47a0eca5ffee33e" -> null

Do you really want to destroy all resources?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

  Enter a value: yes
```

```
aws_launch_template.web: Destruction complete after 1s
aws_lb.internal: Destroying... [id=arn:aws:elasticloadbalancing:eu-north-1:505999908978:loadbalancer/app/lab007-internal-alb/e589738bc0ae16
69]
aws_subnet.public[1]: Destruction complete after 2s
aws_subnet.public[0]: Destruction complete after 2s
aws_autoscaling_group.backend: Destruction complete after 1m59s
aws_launch_template.backend: Destroying... [id=lt-035f87ab10b249547]
aws_launch_template.backend: Destruction complete after 0s
aws_iam_instance_profile.profile: Destroying... [id=lab007-ec2-profile]
aws_security_group.backend: Destroying... [id=sg-0d90eb30357eb979a]
aws_security_group.backend: Destruction complete after 1s
aws_iam_instance_profile.profile: Destruction complete after 2s
aws_iam_role.ec2_role: Destroying... [id=lab007-ec2-ssm-role]
aws_iam_role.ec2_role: Destruction complete after 1s
aws_lb.internal: Still destroying... [id=arn:aws:elasticloadbalancing:eu-north-1...p/lab007-internal-alb/e589738bc0ae1669, 00m10s elapsed]
aws_lb.internal: Still destroying... [id=arn:aws:elasticloadbalancing:eu-north-1...p/lab007-internal-alb/e589738bc0ae1669, 00m20s elapsed]
aws_lb.internal: Destruction complete after 29s
aws_security_group.alb_internal: Destroying... [id=sg-02cf0ccca079c56ac]
aws_subnet.private[0]: Destroying... [id=subnet-0e5348659382faaba]
aws_subnet.private[1]: Destroying... [id=subnet-0d0d918d6471073c0]
aws_subnet.private[0]: Destruction complete after 1s
aws_subnet.private[1]: Destruction complete after 1s
aws_security_group.alb_internal: Destruction complete after 1s
aws_security_group.web: Destroying... [id=sg-04059bae64fda041b]
aws_security_group.web: Destruction complete after 1s
aws_security_group.bastion: Destroying... [id=sg-0c4f6c7b55814420e]
aws_security_group.alb: Destroying... [id=sg-0859b789e99eb0bcc]
aws_security_group.bastion: Destruction complete after 1s
aws_security_group.alb: Destruction complete after 1s
aws_vpc.main: Destroying... [id=vpc-0a47a0eca5ffee33e]
aws_vpc.main: Destruction complete after 2s

Destroy complete! Resources: 37 destroyed.
```